

Problem 1

(a)

We show that $f(x) = \sin(x^2)$ is not uniformly continuous on $[1, \infty)$.

Let $\epsilon = \frac{1}{2}$. We will show that for every $\delta > 0$, there exist $x, y \in [1, \infty)$ such that $|x - y| < \delta$ but $|f(x) - f(y)| \geq \epsilon$.

Let $\delta > 0$ be given. Choose $n \in \mathbb{N}$ large enough so that

$$\frac{\frac{\pi}{2}}{\sqrt{2\pi n + \frac{\pi}{2}} + \sqrt{2\pi n}} < \delta.$$

Define

$$x = \sqrt{2\pi n + \frac{\pi}{2}}, \quad y = \sqrt{2\pi n}.$$

Then $x, y \in [1, \infty)$.

Moreover,

$$\begin{aligned} |x - y| &= \left| \sqrt{2\pi n + \frac{\pi}{2}} - \sqrt{2\pi n} \right| \\ &= \frac{\frac{\pi}{2}}{\sqrt{2\pi n + \frac{\pi}{2}} + \sqrt{2\pi n}} \\ &< \delta. \end{aligned}$$

However,

$$\begin{aligned} |f(x) - f(y)| &= |\sin(x^2) - \sin(y^2)| \\ &= |\sin(2\pi n + \frac{\pi}{2}) - \sin(2\pi n)| \\ &= |1 - 0| = 1 \geq \frac{1}{2}. \end{aligned}$$

Since this holds for every $\delta > 0$, f is not uniformly continuous on $[1, \infty)$.

(b)

Let $\epsilon > 0$ and let $x, y \in [1, \infty)$. Using the identity

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2),$$

we obtain

$$|a - b| = \frac{|a^3 - b^3|}{|a^2 + ab + b^2|}.$$

Let $g(x) = x^{1/3}$. Then

$$\begin{aligned} |g(x) - g(y)| &= |x^{1/3} - y^{1/3}| \\ &= \frac{|x - y|}{x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}}. \end{aligned}$$

Since $x, y \geq 1$, we have

$$x^{2/3} \geq 1, \quad x^{1/3}y^{1/3} \geq 1, \quad y^{2/3} \geq 1,$$

so

$$x^{2/3} + x^{1/3}y^{1/3} + y^{2/3} \geq 3.$$

Thus,

$$|x^{1/3} - y^{1/3}| \leq \frac{|x - y|}{3}.$$

Choose $\delta = 3\epsilon$. Then if $|x - y| < \delta$,

$$|g(x) - g(y)| \leq \frac{|x - y|}{3} < \frac{3\epsilon}{3} = \epsilon.$$

Therefore, g is uniformly continuous on $[1, \infty)$.

Problem 2

(a)

Consider the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x-2)^n}{n \cdot 3^n}.$$

Using the Ratio Test, compute

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (x-2)^{n+1}}{(n+1)3^{n+1}} \cdot \frac{n3^n}{(-1)^n (x-2)^n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-1)(x-2)n}{(n+1)3} \right| \\ &= \frac{|x-2|}{3} \cdot \lim_{n \rightarrow \infty} \frac{n}{n+1} \\ &= \frac{|x-2|}{3}. \end{aligned}$$

By the Ratio Test, the series converges when

$$\frac{|x-2|}{3} < 1 \iff |x-2| < 3.$$

Thus, the radius of convergence is

$$R = 3.$$

(b) The center is $a = 2$, so

$$|x-2| < 3 \iff -1 < x < 5.$$

At $x = -1$:

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{(-1)^n (-3)^n}{n \cdot 3^n} &= \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^n 3^n}{n \cdot 3^n} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n} \\ &= \sum_{n=1}^{\infty} \frac{1}{n}. \end{aligned}$$

This is a p -series with $p = 1$. Since $p \leq 1$, the series diverges by the p -series test. Hence, the series diverges at $x = -1$.

At $x = 5$:

$$\sum_{n=1}^{\infty} \frac{(-1)^n (3)^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}.$$

This is an alternating series. Let $a_n = \frac{1}{n}$. Then:

- $a_n \geq 0$ for all n ,
- a_n is decreasing since $\frac{1}{n} \geq \frac{1}{n+1}$,
- $\lim_{n \rightarrow \infty} a_n = 0$.

Therefore, by the Alternating Series Test, the series converges at $x = 5$.

Hence, the interval of convergence is

$$(-1, 5].$$

Problem 3

(a)

Let $\epsilon > 0$. For all $x \in [0, \infty)$,

$$|f_n(x) - f(x)| = \left| \left(2x + \frac{1}{\sqrt{n}} \right) - 2x \right| = \frac{1}{\sqrt{n}}.$$

Choose $N \in \mathbb{N}$ such that

$$\frac{1}{\sqrt{N}} < \epsilon, \quad \text{for example } N = \left\lceil \frac{1}{\epsilon^2} \right\rceil.$$

Then for all $n \geq N$ and all $x \in [0, \infty)$,

$$|f_n(x) - f(x)| = \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \epsilon.$$

Therefore, $f_n \rightarrow f$ uniformly on $[0, \infty)$.

(b)

Let $g_n(x) = \left(2x + \frac{1}{\sqrt{n}} \right)^2$ and $g(x) = 4x^2$.

Let $x \in [0, \infty)$ be fixed and let $\epsilon > 0$. We show that there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|g_n(x) - g(x)| < \epsilon.$$

Compute:

$$\begin{aligned} |g_n(x) - g(x)| &= \left| \left(2x + \frac{1}{\sqrt{n}} \right)^2 - 4x^2 \right| \\ &= \frac{4x}{\sqrt{n}} + \frac{1}{n}. \end{aligned}$$

Choose

$$N_1 = \left\lceil \left(\frac{8x}{\epsilon} \right)^2 \right\rceil, \quad N_2 = \left\lceil \frac{2}{\epsilon} \right\rceil,$$

and let

$$N = \max\{N_1, N_2\}.$$

Then for all $n \geq N$, we have

$$\frac{4x}{\sqrt{n}} \leq \frac{4x}{\sqrt{N_1}} \leq \frac{\epsilon}{2}, \quad \frac{1}{n} \leq \frac{1}{N_2} \leq \frac{\epsilon}{2}.$$

Thus,

$$|g_n(x) - g(x)| = \frac{4x}{\sqrt{n}} + \frac{1}{n} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, $g_n(x) \rightarrow g(x)$ for each fixed $x \in [0, \infty)$, so $g_n \rightarrow g$ pointwise.

(c)

We show that g_n does not converge uniformly to g on $[0, \infty)$.

Let $\epsilon = 1$. For any $n \in \mathbb{N}$, choose

$$x_n = \frac{\sqrt{n}}{4}.$$

Then

$$\begin{aligned} |g_n(x_n) - g(x_n)| &= \frac{4x_n}{\sqrt{n}} + \frac{1}{n} \\ &= \frac{4(\sqrt{n}/4)}{\sqrt{n}} + \frac{1}{n} \\ &= 1 + \frac{1}{n} > 1 = \epsilon. \end{aligned}$$

Thus, for every n , there exists $x_n \in [0, \infty)$ such that

$$|g_n(x_n) - g(x_n)| \geq \epsilon.$$

Therefore, g_n does not converge uniformly to g on $[0, \infty)$.

Problem 4

(a)

Case 1: $x = 0$. Then

$$f_n(0) = n \cdot 0 \cdot e^{-n^2 \cdot 0^2} = 0 \quad \text{for all } n,$$

so clearly

$$\lim_{n \rightarrow \infty} f_n(0) = 0.$$

Case 2: $x \neq 0$. Write

$$f_n(x) = nxe^{-(nx)^2}.$$

Let $y = nx$. Then as $n \rightarrow \infty$, $y \rightarrow \infty$, and

$$f_n(x) = ye^{-y^2}.$$

Now

$$\lim_{y \rightarrow \infty} ye^{-y^2} = \lim_{y \rightarrow \infty} \frac{y}{e^{y^2}} = 0$$

by standard exponential growth.

Therefore, the pointwise limit is

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0 \quad \text{for all } x \in \mathbb{R}.$$

(b)

Compute the derivative of f_n :

$$f'_n(x) = \frac{d}{dx}(nxe^{-n^2x^2}) = ne^{-n^2x^2} + nx \cdot (-2n^2x)e^{-n^2x^2} = ne^{-n^2x^2}(1 - 2n^2x^2).$$

Set $f'_n(x) = 0$ to find critical points:

$$1 - 2n^2x^2 = 0 \quad \implies \quad x = \frac{1}{\sqrt{2n}}.$$

At this x , f_n attains its maximum. Evaluate:

$$f_n\left(\frac{1}{\sqrt{2n}}\right) = n \cdot \frac{1}{\sqrt{2n}} \cdot e^{-n^2 \cdot (1/2n^2)} = \frac{1}{\sqrt{2}} e^{-1/2} \approx 0.43.$$

Since

$$\sup_{x \in [0, \epsilon]} |f_n(x) - f(x)| \geq f_n\left(\frac{1}{\sqrt{2n}}\right) \not\rightarrow 0,$$

it follows that f_n does not converge uniformly to f on any interval containing 0.

(c)

For $x \geq a > 0$, observe

$$0 \leq f_n(x) = nxe^{-n^2x^2} \leq nxe^{-n^2a^2} \leq ne^{-n^2a^2} \cdot x_{\max} \leq ne^{-n^2a^2} \cdot x,$$

but the key point is that $e^{-n^2 a^2}$ decays faster than any polynomial in n . In particular,

$$\lim_{n \rightarrow \infty} n e^{-n^2 a^2} = 0.$$

Therefore, for all $x \in [a, \infty)$,

$$|f_n(x) - f(x)| = |f_n(x)| \leq n x e^{-n^2 a^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since the bound is independent of x , it follows that

$$\sup_{x \in [a, \infty)} |f_n(x) - f(x)| \rightarrow 0,$$

and hence $f_n \rightarrow f$ uniformly on $[a, \infty)$ for any $a > 0$.

Problem 5

(a)

Consider $f_n(x) = \frac{x^n}{n}$ on $[0, 1]$.

For $x \in [0, 1)$, we have $x^n \rightarrow 0$ as $n \rightarrow \infty$, hence

$$f_n(x) = \frac{x^n}{n} \rightarrow 0.$$

For $x = 1$,

$$f_n(1) = \frac{1}{n} \rightarrow 0.$$

Thus, the pointwise limit is

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0 \quad \text{for all } x \in [0, 1].$$

(b)

We compute

$$\sup_{x \in [0, 1]} |f_n(x) - f(x)| = \sup_{x \in [0, 1]} \frac{x^n}{n}.$$

Since $0 \leq x \leq 1$, the maximum of x^n occurs at $x = 1$, so

$$\sup_{x \in [0, 1]} \frac{x^n}{n} = \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, it follows that

$$\sup_{x \in [0, 1]} |f_n(x) - f(x)| \rightarrow 0.$$

Therefore, $f_n \rightarrow f$ uniformly on $[0, 1]$.

(c)

Let $h_n(x) = f'_n(x)$. Then

$$h_n(x) = \frac{d}{dx} \left(\frac{x^n}{n} \right) = x^{n-1}.$$

(i)

For $x \in [0, 1)$, we have $x^{n-1} \rightarrow 0$, and for $x = 1$,

$$h_n(1) = 1^{n-1} = 1.$$

Thus, the pointwise limit is

$$h(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

(ii)

Each h_n is continuous on $[0, 1]$ because it is differentiable. However, the limit function h is not continuous at $x = 1$.

Since the uniform limit of continuous functions must be continuous, it follows that h_n does not converge uniformly to h on $[0, 1]$.